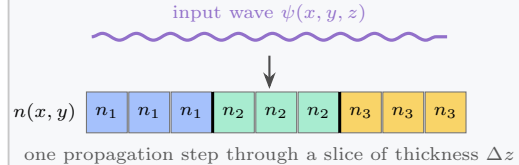


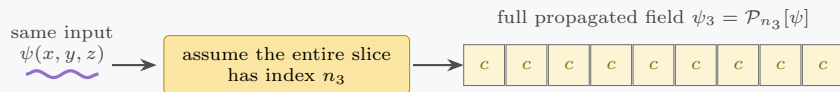
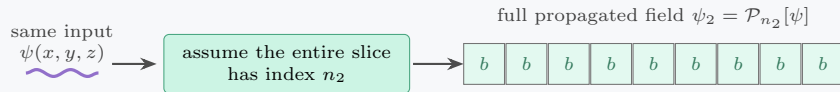
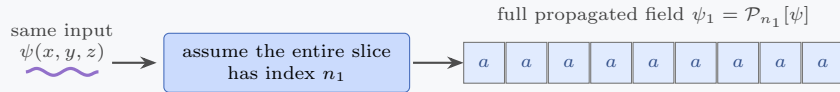
1. Partition the heterogeneous slice



Each binary mask selects the output pixels whose local index is n_m :

$$M_m(x, y) = \begin{cases} 1, & n(x, y) = n_m, \\ 0, & \text{otherwise.} \end{cases}$$

2. Propagate the full wave for each refractive index



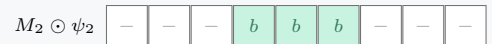
For every representative index, apply its homogeneous wide-angle propagator over the complete transverse grid:

$$\psi_m = \mathcal{F}^{-1} \left[e^{i\Delta z \sqrt{k_0^2 n_m^2 - k_\perp^2}} \mathcal{F}\{\psi(x, y, z)\} \right].$$

3. Apply masks and assemble



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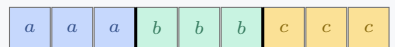


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next wave plane $\psi(x, y, z + \Delta z)$



$$\psi(x, y, z + \Delta z) = \sum_{m=1}^3 M_m(x, y) \odot \psi_m(x, y).$$